

Path Independency of Thermodynamics State Properties and the Ideal Gas Law

CHE260 Lab Report by Rooney Cheung, Jady Zeng, Akhilesh Menon, Session PRA0101.

Abstract

This classical thermodynamics experiment had two primary objectives: (1) demonstrate the path independence of thermodynamic state properties, and (2) determine the volume ratio between two interconnected gas tanks and the initial mass and volume of one tank using the Ideal Gas Law. Air was expanded from an initially pressurized tank ($P_{G,1} \approx 40$ psi) into an evacuated tank ($P_{G,2} \approx -6$ psi) using two distinct paths: a rapid expansion (using a solenoid valve) and a slow, gradual expansion (using a micrometer needle valve). Comparing the final pressures and temperatures confirmed that the final state is independent of the expansion path. The initial mass of air added to the left tank was then determined by integrating the mass flow rate over time, and the tank volume was determined using the measured final pressure and temperature to the Ideal Gas Law ($PV=mRT$), yielding $(5.5 \pm 0.7) \cdot 10^{-3} \text{m}^3$. The calculated volume ratios were consistent across both expansion paths ($\%V_a = 0.56 \pm 0.09$ and $\%V_b = 0.55 \pm 0.09$), and the final state properties were found to be equal within experimental uncertainty, successfully demonstrating path independence.

Introduction

This lab report examines the volume ratio of air between two connected tanks and evaluates thermodynamic predictions, with a focus on demonstrating the path independence [1] of final state properties and determining the initial mass and volume of one of the tanks.

Pressure was equalized rapidly between a 40psi tank and -6psi tank, where in one case thermal and pressure equilibrium is reached gradually, and the second case rapidly instead. Understanding the behaviour of gases during expansion is fundamental in important engineering systems such as pneumatic systems and energy storage devices [2], and reinforces the concept that state properties depend on the current state and not the path taken.

The experiment aims to (1) calculate the volume ratio between the two tanks, (2) discuss the heat transfer throughout the experiment, (3) demonstrate path independency of state properties, and (4) determine the initial mass and volume of air of the 40psi (left) tank. By analyzing the experimentally obtained mass-flow rate, pressure-time, and temperature-time data, and applying the ideal gas law and the conservation of mass, the two tank's volume ratio and initial mass and volume of air in the 40psi tank can be determined, while also allowing for the analysis of the path independence of the system's state properties.

Experimental Method

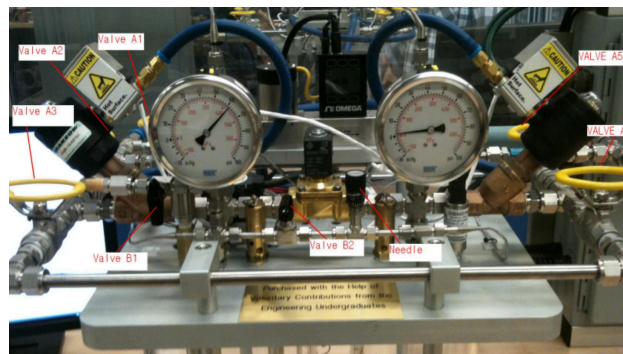


Figure 1: Experimental Apparatus with labelled valves [4]

Part 1: Path Independency and Volume Ratio Determination

Initial Setup (For both Method 1 and Method 2):

Record the ambient atmospheric pressure (P_{atm}) with an external manometer and start collecting data using the LabVIEW program. In order to pressurize the left tank (T_1, P_1),

open the A2 valve, and using LabVIEW open the Left Solenoid Valve and set the mass flowrate to 50g/min. Once the left tank pressure (P1) reaches 40 psi as shown on the graph, close the Left Solenoid Valve, the A2 valve, and set the mass flowrate to 0g/min. Then vacuum the right tank (T2, P2) by opening the A1 and A5 valves. Using LabVIEW, open the Right Solenoid Valve and set the mass flowrate to 50 g/min. When the right tank pressure reaches -6 psi, close the Right Solenoid Valve, then the A1 and A5 valves, and set the flow rate to 0g/min.

Two approaches are used compare the effect of the process path on the final state properties:

- ***Method 1 - Rapid Expansion (Solenoid Valve):*** After completing the initial setup, open the Center Solenoid Valve using LabVIEW. Once the temperature difference between the two tanks becomes insignificant (approximately 45 second wait), stop data collection and save the data as Lab 2 - Part 1a. Reset the LabVIEW program, close all valves, and empty both tanks.
- ***Method 2 - Slow Expansion (Micrometer Needle Valve):*** After completing the initial setup, open the Small Ball Valve (B2) and the micrometer needle valve by turning it 4 times counterclockwise. Once the pressure difference between tanks becomes insignificant (approximately 10 minute wait), stop data collection and save data as **Lab2 - Part1b**. Reset the LabVIEW program, close all valves, and empty both tanks.

Part 2: Determining Initial Mass and Volume of the Left Tank

Record the ambient pressure using an external manometer and start data collection in the LabVIEW program. Pressurize and stabilize the left tank by opening the A2 valve, and using LabVIEW open the Left Solenoid Valve and set the mass flowrate to 50g/min. Once the left tank pressure (P1) reaches 40 psi as shown on the graph, close the Left Solenoid Valve, and

wait for P1 and T1 to remain constant with time. Stop data collection and save data as **Lab 2**

- **Part2a.** Reset the LabVIEW program, close all valves, and empty both tanks.

Results and Discussion

[1] Path Independency of Thermodynamics State Properties Volume Ratio of the Tanks

During the experiment, values of Pressure, P , Temperature, T , are collected for each tank (left = 1, right = 2). The results of Part 1a are collected and plotted against time as shown below:

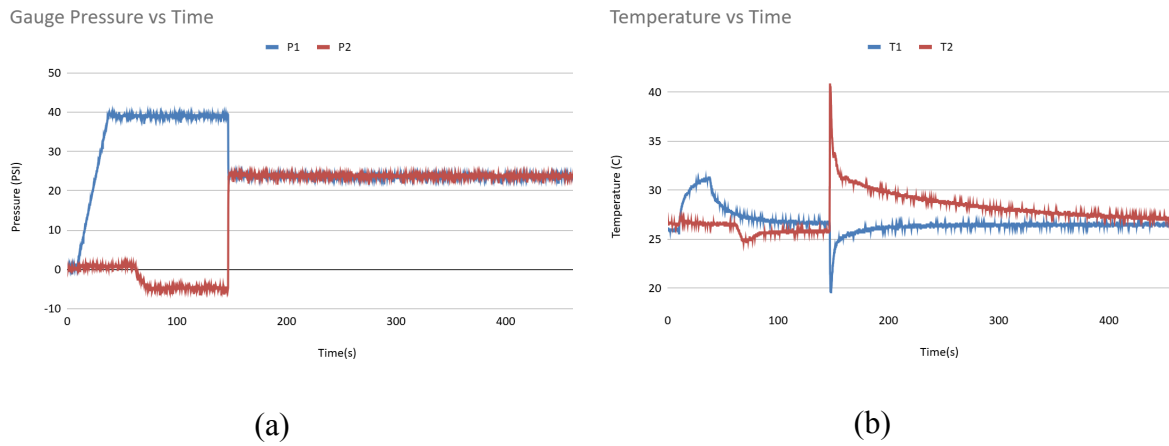


Figure 2: Pressure (a) and Temperature (b) of connected tanks as a function of time. Tanks are brought to equilibrium at $t=500s$.

Refer to Figure 2a, Gauge pressure at $t \approx 150s$ were $P_{G1} = 39.5 \pm 1.0$ psi and $P_{G2} = -5.0 \pm 1.0$ psi respectively. At $t \approx 500s$, the equilibrium gauge pressure was $P_{G,eq} = 23.6 \pm 1.4$ psi.

Refer to Figure 2b, Temperature at $t \approx 150s$ were $T_1 = 26.8 \pm 0.5$ C° and $T_2 = 25.8 \pm 0.5$ C° respectively, the equilibrium at $t \approx 500s$, the equilibrium temperature was $T_{eq} = 27.0 \pm 0.7$ C°.

Note particularly the rapid process indicated by the steep slopes at $t \approx 150s$ for both figures.

The results of Part 1b are collected and plotted against time as shown below:

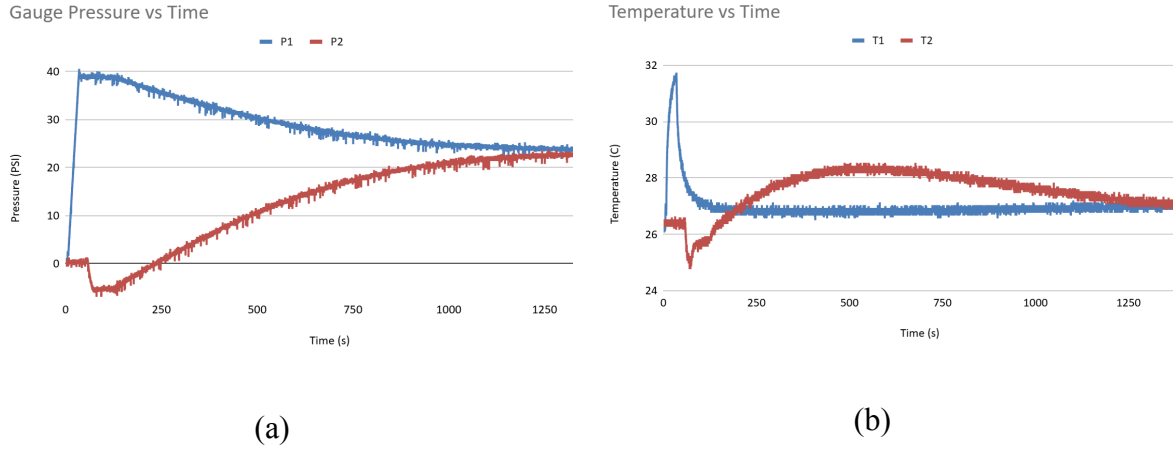


Figure 3: Pressure (a) and temperature (b) of connected tanks as a function of time. As the B2 valve opens, both tanks' pressure converge to equilibrium (23.8 psi, 27.1 C°).

Refer to Figure 3a, Gauge pressure at $t \approx 150$ s were $P_{G1} = 39.5 \pm 1.0$ psi and $P_{G2} = -5.0 \pm 1.0$ psi respectively. At $t \approx 1380$ s, the equilibrium gauge pressure was $P_{G,eq} = 23.8 \pm 1.4$.

Refer to Figure 3b, Temperature at $t \approx 150$ s were $T_1 = 27.0 \pm 0.2$ C° and $T_2 = 26.2 \pm 0.2$ C° respectively. At $t \approx 500$ s, the equilibrium temperature was $T_{eq} = 27.1 \pm 0.3$ C°. Note particularly the gradual process indicated by the relatively plain slopes throughout.

The uncertainties of all initial values are estimated by the fluctuation of data present in the chart, and the uncertainties of equilibrium values are propagated from initial uncertainties, These are the measurement errors arising from the manometers and thermostats.

Using the Ideal Gas Law, and the Principle of Conservation of Mass, and conversion factors, the following relation between volume ratio can be deduced,

$$\text{Ideal Gas Law: } PV = mRT, m = \frac{PV}{RT}, V = \frac{mRT}{P} - [1]$$

Principle of Conservation of Mass: $\sum_{k=1}^n m_{i,k} = \sum_{k=1}^n m_{f,k} - [2]$

Absolute Pressure, $P_A = P_{Atm} + P_G$; Absolute Temperature, $T_K = T_C + 273.15K - [3]$

($P_{Atm} = 30.09 \pm 0.1 \text{ inHg} = 14.78 \pm 0.05 \text{ psi}$)

Ratio of Volumes and its uncertainty, $\%V = \frac{V_2}{V_1} = \left(\frac{P_{A,1}}{T_{K,1}} - \frac{P_{A,eq}}{T_{K,eq}} \right) \div \left(\frac{P_{A,eq}}{T_{K,eq}} - \frac{P_{A,2}}{T_{K,2}} \right) - [4]$

$\delta\%V = \sqrt{\left(\left(\frac{\delta P_{A,1}}{P_{A,1}} \right)^2 + 2 \left(\frac{\delta T_{K,1}}{T_{K,1}} \right)^2 + \left(\frac{\delta P_{A,eq}}{P_{A,eq}} \right)^2 \right) \div \left(\left(\frac{\delta P_{A,2}}{P_{A,2}} \right)^2 + 2 \left(\frac{\delta T_{K,2}}{T_{K,2}} \right)^2 + \left(\frac{\delta P_{A,eq}}{P_{A,eq}} \right)^2 \right)} - [5]$

The volume ratio for Part 1a, $\%V_a = 0.56 \pm 0.09$; for Part 1b, $\%V_b = 0.55 \pm 0.09$

The two volume ratios are approximately equal, with negligible differences, the average,

$\%V_{avg} = 0.56 \pm 0.13$

Heat transfer is the energy transfer from a region of higher temperature to a region of lower temperature [3]. During both parts of the experiment, heat transfer occurs between the tanks and the surroundings, in this case the air of the room ($T_{atm} = 25.0^\circ\text{C}$ on the experiment day).

Neglecting short periods of abrupt change, T_1 , T_2 is generally greater than T_{atm} . When we consider the tanks as the system and the atmosphere as the surroundings, we conclude that a net heat transfer must occur from our system (tanks) to the surroundings (the room's air).

Heat loss is another error factor in this experiment, as an isolated system cannot be implemented, the equilibrium temperature approaches the room temperature indefinitely.

When considering only the initial and equilibrium values, the initial values for both experiments have negligible differences, and the difference between the two equilibrium temperatures and pressures from Part 1a and 1b, ΔT_{eq} and ΔP_{eq} , is much smaller ($\approx < 15\%$) than

their respective propagated uncertainties. As such, we conclude that the equilibrium pressure and temperature values are equal in both Part 1a and 1b within experimental uncertainties.

We can further conclude that the state functions, temperature and pressure are path independent, i.e. for a fixed mass and volume, the final pressure and temperature must be fixed in spite of the different paths taken, if given enough time for the process to propagate. Figure 3 illustrates the idea of path independence, this agrees with the idea of temperature, pressure being path independent in classical thermodynamics [1].

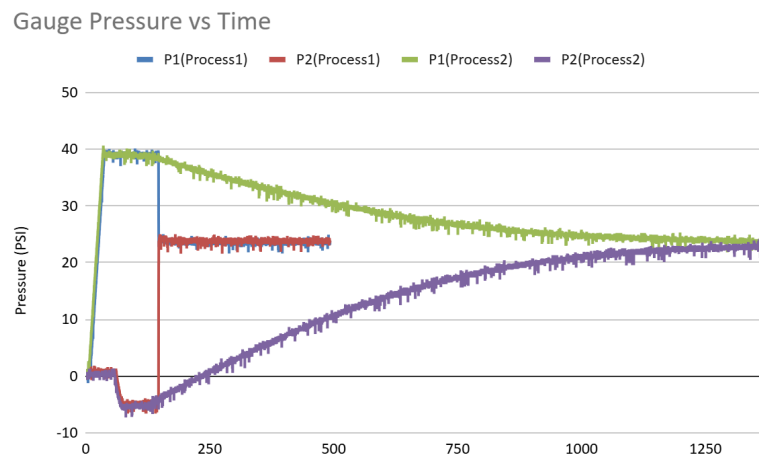


Figure 4: Combined chart for Pressure vs time plotted for Part 1a (Process1) and Part 1b (Process2) to illustrate the idea of path independence.

[2] Determine the Initial Mass and Volume of the Left Tank (40psi)

The experiment produced measurements for mass flow rate, pressure, and temperature over time for the left tank pressurized to 40psi. These values are plotted as a function of time in Figure 4:

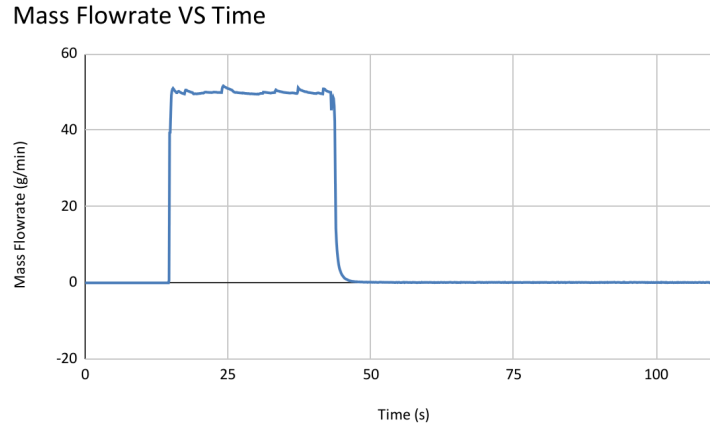


Figure 4: Mass Flowrate (g/min) of the left tank as a function of time, exhibiting a square pulse between $t \sim 15s$ to $t \sim 49s$ with mass flowrate 42.6 g/min

In Figure 4 the mass flowrate dm exhibits a square pulse, peaking suddenly and remaining essentially constant at $42.6 \pm 6 \text{ g/min}$ from $14.7 \pm 0.05s$ to $48.6 \pm 0.05s$, ($\Delta t = 33.9 \pm 0.07s$). The graph remains approximately constant throughout this interval, with small, negligible variations likely due to experimental uncertainty, Thus, the square pulse shape's area, representing the net mass of the system, is determined by:

Mass of air and its uncertainty, $m = dm\Delta t - [6]$

$$\delta m = m \sqrt{\left(\frac{\delta dm}{dm}\right)^2 + \left(\frac{\delta \Delta t}{\Delta t}\right)^2} - [7]$$

This gives the result, $m = 24.2 \pm 3g$

Systematic error arises from the precision of the measurement software, with time accurate to 0.1s. The change in time uncertainty is propagated from the initial measured values.

However, the mass flowrate quantity is determined by averaging the peak values, therefore an accurate representation of its uncertainty is to consider the greatest variation value, which deviates by 6g/min.

The tank's initial volume can be calculated using pressure-time and temperature-time graphs shown below:

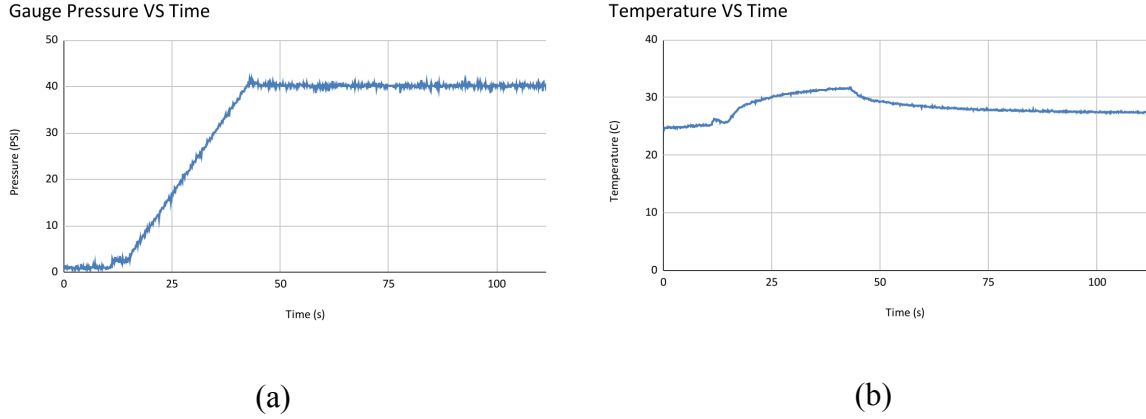


Figure 5: Pressure (a) and temperature (b) of the left tank as a function of time, remaining constant at 40.20 psi 28.18 C° and when valve closes ($t \approx 43$).

Averaging the constant values in Figure 5 from $t \approx 43$ to 110s yields temperature $T = 28.18 \pm 0.5$ C° (301.33 ± 0.5 K), and gauge pressure $P_G = 40.20 \pm 1.0$ psi (277.17 ± 6.89 kPa). Thus, using the Ideal Gas Law (Equation 1), and conversion factors in Equations 3, $R = 0.2870$ kJ/kgK for air, we get

$V = (5.5 \pm 0.7) 10^{-3} \text{m}^3$, and its associated uncertainty,

$$\delta V = V \sqrt{\left(\frac{\delta m}{m}\right)^2 + \left(\frac{\delta T}{T}\right)^2 + \left(\frac{\delta P}{P}\right)^2} - [8]$$

Alike to the mass flowrate measurement, the pressure and temperature were determined through averaging the fluctuating values when it becomes seemingly constant. The pressure does not vary more than 1.0psi from the averaged value, and the temperature does not vary more than 0.5 C°. These fluctuations can be a result of offset instrumentation calibration of the temperature sensor and manometer. Inaccurate placements, time lags, and the assumption

that the tank is an isolated system (no heat/pressure loss) could be a result of the changing measurements when theoretically it should remain constant.

Moreover, the assumption of ideal gas behaviour (incompressibility) influences the mass flowrate measurements. With a lower density, the mass flowrate is underestimated than what is demonstrated. Additionally, compressibility causes the calculated volume to be further underestimated as it is less than the total volume of the chamber [5].

Conclusion

The experimental results successfully demonstrated two key principles of thermodynamics and gas behavior. First, the ratio of volumes between the two tanks, calculated using the Ideal Gas Law and conservation of mass, was consistent across both the rapid expansion

(Method 1: $\%V_a = 0.56 \pm 0.09$) and the slow expansion (Method 2: $\%V_b = 0.55 \pm 0.09$) processes. More critically, by comparing the final equilibrium state properties, the experiment confirmed the principle of path independence: despite taking vastly different paths to equilibrium (a sudden, rapid equalization versus a slow, gradual one), the final equilibrium pressure and temperature were found to be equal within the calculated experimental uncertainties. This reinforces the core thermodynamic concept that state properties depend only on the initial and final states, not the process taken between them. Finally, analysis of the mass flow rate data determined the initial mass of air added to the left tank to be

$m = 24.2 \pm 3 \text{ g}$, which was then used to calculate the tank's volume as $V = (5.5 \pm 0.7) \times 10^{-3} \text{ m}^3$

at the measured final conditions.

Citations

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